

The Literary Digest: A Story about Sampling

MATH 213

Group 1

Literary Digest was a weekly political magazine founded in 1890. It was something like today's Time magazine. Starting in 1916, the magazine made predictions of the winner in US presidential elections, using different sorts of polls. The predictions were correct for the five presidential elections between 1916 and 1932.

By 1932, Literary Digest had a method that seemed to work very well: they sent out millions of straw-vote ballots to people on a large mailing list. They then published a summary of the votes from those people who chose to send their ballots back. In 1932, they predicted the percentage of the popular vote for the winner (Franklin Roosevelt) to within one percentage point.

In August 1936, Literary Digest sent out about 10 million such ballots, using a list drawn mostly from telephone books and automobile registration records. (In developing their list, they were thinking, in part, about contacting people who might subscribe to Literary Digest: every ballot came with advertising material about the magazine.)

By the end of October 1936, about 2.4 million people had responded, with 54% picking Alf Landon (the Republican) and 41% picking Franklin Roosevelt (the Democrat).

1. Identify the following parts of this study

Population of interest: American population

Sample: 10 million randomly generated from phonebook.

Variable of interest: Voting choice

Type of variable (categorical or numerical): categorical

Parameter(s) of interest: percentage of party

Statistic(s): percent of people who vote on either side.

2. When the actual election occurred on November 3rd, Franklin Roosevelt won 61% of the popular vote, and Alf Landon won 37%.

Give two or three possible explanations for why the Literary Digest prediction was so far off. Think, in particular, about *bias* in the sampling method: why might the design of this poll have led to an overestimate of the vote for Landon and an underestimate of the vote for Roosevelt?

Half of their sample not replying was probably a large part of it. There was no way to ensure a diversity of backgrounds and views. Also, the fact that advertising came with the ballot probably didn't help. They also very clearly prioritized people of more wealth, judging by the fact that they made the list based off of automobile and phone records, so they were likely missing a large part of the population below that wealth gap.

Group 2

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By the end of October 1936, about 2.4 million people had responded, with 54% picking Alf Landon (the Republican) and 41% picking Franklin Roosevelt (the Democrat).

1. Identify the following parts of this study

Population of interest: The voting population

Sample: People on the mailing list who sent their ballot back

Variable of interest: Voting democrat or republican

Type of variable (categorical or numerical): categorical nominal

Parameter(s) of interest: Percentage of the population who vote democrat/republican

Statistic(s): Percent of people who vote either side in the sample

2. When the actual election occurred on November 3rd, Franklin Roosevelt won 61% of the popular vote, and Alf Landon won 37%.

Give two or three possible explanations for why the Literary Digest prediction was so far off. Think, in particular, about *bias* in the sampling method: why might the design of this poll have led to an overestimate of the vote for Landon and an underestimate of the vote for Roosevelt?

1. They in part based who to send the ballots to based on who they thought may be more likely to subscribe to the Literary Digest magazine, looking for people who were in automobile records or telephone books and including promotional material. IE, “Random sample” was not truly random
2. Republican voters may have been more likely to send their straw-vote ballot back than democratic voters.

Group 3

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1. Identify the following parts of this study

Population of interest: US voting population

Sample: 10 million straw-vote ballots

Variable of interest: Presidential Vote

Type of variable (categorical or numerical): Categorical

Parameter(s) of interest: Proportion of Roosevelt voters to Alf voters

Statistic(s): 54% picking Alf Landon (the Republican) and 41% picking Franklin Roosevelt (the Democrat)

2. When the actual election occurred on November 3rd, Franklin Roosevelt won 61% of the popular vote, and Alf Landon won 37%.

Give two or three possible explanations for why the Literary Digest prediction was so far off. Think, in particular, about *bias* in the sampling method: why might the design of this poll have led to an overestimate of the vote for Landon and an underestimate of the vote for Roosevelt?

Polls were not returned

They sent polls to their subscribers

Group 4

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1. Identify the following parts of this study

Population of interest: Americans

Sample: Those on the mailing list

Variable of interest: Vote

Type of variable (categorical or numerical): Categorical

Parameter(s) of interest: The percentage of Democratic or Republican votes in the population

Statistic(s): The percentage of Democratic or Republican votes in the sample

2. When the actual election occurred on November 3rd, Franklin Roosevelt won 61% of the popular vote, and Alf Landon won 37%.

Give two or three possible explanations for why the Literary Digest prediction was so far off. Think, in particular, about *bias* in the sampling method: why might the design of this poll have led to an overestimate of the vote for Landon and an underestimate of the vote for Roosevelt?

1. The people who own cars might have been mostly Republican
2. Not everyone responded (only ~20% of people responded)
3. Ballots were sent to people who were likely to subscribe/Some people may have thought it was an ad

Group 5

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1. Identify the following parts of this study

Population of interest: The US population that can vote

Sample: People that received ballots from Literary Digest

Variable of interest: Presidential vote

Type of variable (categorical or numerical): Categorical

Parameter(s) of interest: The percentage of votes for each president in the election

Statistic(s): The percentage of votes for each president from the people who were mailed Literary Digest's ballot

2. When the actual election occurred on November 3rd, Franklin Roosevelt won 61% of the popular vote, and Alf Landon won 37%.

Give two or three possible explanations for why the Literary Digest prediction was so far off. Think, in particular, about *bias* in the sampling method: why might the design of this poll have led to an overestimate of the vote for Landon and an underestimate of the vote for Roosevelt?

The Literary Digest chose people in the sample who they thought might subscribe to their magazine which could lead to unintentional bias because people of a certain political party might be more inclined to subscribe.

They also only chose drawings from car registrations and phone books which may have also led to sampling bias since it wasn't 100% random.

Group 6

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1. Identify the following parts of this study

Population of interest: Voting Americans

Sample: People on mailing list

Variable of interest: Who they vote for

Type of variable (categorical or numerical): categorical

Parameter(s) of interest: Percentage of people that vote Democrat vs Republican

Statistic(s): Percentage of people on mailing list who voted Democrat vs Republican

2. When the actual election occurred on November 3rd, Franklin Roosevelt won 61% of the popular vote, and Alf Landon won 37%.

Give two or three possible explanations for why the Literary Digest prediction was so far off. Think, in particular, about *bias* in the sampling method: why might the design of this poll have led to an overestimate of the vote for Landon and an underestimate of the vote for Roosevelt?

- Not the entire population of Americans that are eligible to vote
- The magazine could be more popular with a specific political party

Group 7

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1. Identify the following parts of this study

Population of interest: All Election Voters

Sample: Responders to Literary digest ballot mail.
2.4 million

Variable of interest: Who are you voting for

Type of variable (categorical or numerical): Categorical

Parameter(s) of interest: Percentage of voters in the country

Statistic(s): 54% Landon

2. When the actual election occurred on November 3rd, Franklin Roosevelt won 61% of the popular vote, and Alf Landon won 37%.

Give two or three possible explanations for why the Literary Digest prediction was so far off. Think, in particular, about *bias* in the sampling method: why might the design of this poll have led to an overestimate of the vote for Landon and an underestimate of the vote for Roosevelt?

- The subscribers of Liberty Digest may of had a heavy population of republicans in their audience.
- The republicans may have been more eager to had their voice heard.

Group 8

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By the end of October 1936, about 2.4 million people had responded, with 54% picking Alf Landon (the Republican) and 41% picking Franklin Roosevelt (the Democrat).

1. Identify the following parts of this study

Population of interest:

American citizens eligible to vote

Sample:

Americans willing/able to vote and whose name's are present in automobile records or telephone books

Variable of interest:

The voters desired candidate

Type of variable (categorical or numerical): categorical

Parameter(s) of interest:

Percentage of the total American voting population that voted for a certain candidate

Statistic(s):

Number/percentage of people surveyed who voted for a certain candidate

2. When the actual election occurred on November 3rd, Franklin Roosevelt won 61% of the popular vote, and Alf Landon won 37%.

Give two or three possible explanations for why the Literary Digest prediction was so far off. Think, in particular, about *bias* in the sampling method: why might the design of this poll have led to an overestimate of the vote for Landon and an underestimate of the vote for Roosevelt?

One possible explanation for why the sampling was so off is that there could have been a social stigma against voting for roosevelt and people at the time were hesitant to admit that they would vote roosevelt.

Another large reason that the sample could be considered biased is who they chose to survey. In the year 1936, America was in the midst of the Great Depression, yet Literary Digest chose to mail their polls based off of phone number lists and automobile registry, which were commodities not everyone had at the time. This essentially catered the surveys to the upper class of the American populace at the time, and as such was not representative of the true population.

Group 9

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By the end of October 1936, about 2.4 million people had responded, with 54% picking Alf Landon (the Republican) and 41% picking Franklin Roosevelt (the Democrat).

1. Identify the following parts of this study

Population of interest: general public with voting rights

Sample: people who sent back their ballots (2.4 million)

Variable of interest: candidate/party

Type of variable (categorical or numerical): categorical

Parameter(s) of interest: percentage of votes received for [candidate] (like Roosevelt) in the population

Statistic(s): percentage of votes received for [candidate] (like Roosevelt) in the sample

2. When the actual election occurred on November 3rd, Franklin Roosevelt won 61% of the popular vote, and Alf Landon won 37%.

Give two or three possible explanations for why the Literary Digest prediction was so far off. Think, in particular, about *bias* in the sampling method: why might the design of this poll have led to an overestimate of the vote for Landon and an underestimate of the vote for Roosevelt?

Based upon people who they thought read their magazine or would be interested in subscribing

Used phone and car data to send to population

Migrant voters, people working somewhere else than they live, would not necessarily respond to survey

Group 10

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By the end of October 1936, about 2.4 million people had responded, with 54% picking Alf Landon (the Republican) and 41% picking Franklin Roosevelt (the Democrat).

1. Identify the following parts of this study

Population of interest: American Voters

Sample: 2.4 million that responded (1936 Election)

Variable of interest: Candidate Selected

Type of variable (categorical or numerical): Categorical

Parameter(s) of interest: “actual election occurred on November 3rd, Franklin Roosevelt won 61% of the popular vote, and Alf Landon won 37%.”

They have a phone and car so they are upper class rather than average american for sampling purposes

Statistic(s): 54% picking Alf Landon (the Republican) and 41% picking Franklin Roosevelt (the Democrat).

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